Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Virtual Reality–Based Game-Oriented Ubiquitous Worship Learning Technology for Individuals with Developmental Disabilities**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information and Communication			
	Medium Category	Small Category	Medium Category	Small Category
	Digital Content	*Visual Reality*		
Concept of Proposed Technology	**Core Concept** 1) VR-Based Visual Learning Technologies ● Implementation of 3D virtual environments that replicate real worship spaces ● Enhancement of predictability through visually centered information delivery 2) Game-Based Learning Design Technologies ● Structuring worship procedures into step-by-step missions and interactive elements ● Application of feedback and reward mechanisms to encourage repeated participation 3) Ubiquitous Learning (u-Learning) Environment Technologies ● Provision of learning access without constraints of time and location ● Support for multiple device platforms **Key Functions**			

	1) Usability ● Intuitive interfaces designed for individuals with developmental disabilities 2) Minimal operational steps ● Visually oriented designs understandable by users with limited literacy 3) Accessibility ● User interfaces that account for sensory, cognitive, and linguistic abilities 4) Stability & Safety ● Minimization of sensory discomfort and fatigue during VR use 5) Scalability ● Expandability to diverse religious contexts and content domains
Rationale for Joint R&D Project	**Critical Technical Barriers** International research experience and standards related to communication-centered accessibility design for individuals with developmental disabilities and the application of virtual reality technologies have already been accumulated in several advanced countries. In regions such as North America and Europe, numerous empirical studies and practical cases exist in areas including accessibility-oriented design, AAC user experience, and VR-based educational content development. Sharing these experiences and design philosophies through international joint research is essential for reducing trial-and-error and enhancing the overall maturity and quality of the proposed technology. **Why Global Collaboration?** Worship and religious activities are domains that exhibit strong national, cultural, and religious specificity, and design approaches centered on a single country inherently face structural limitations in applicability. Through international collaborative research, comparative analysis of worship structures and communication practices across diverse religious and cultural contexts can be conducted, enabling the development of a universal accessibility model that is not biased toward any specific religion or culture. In the case of VR-based learning technologies, differences exist across countries in terms of hardware environments, software platforms, and user interface design ecosystems. International collaboration allows validation of applicability across diverse device environments and technological infrastructures, thereby ensuring the generality and scalability of the proposed technology. **Expected Impact** Religious participation and accessibility for individuals with developmental disabilities are internationally recognized issues closely linked to the rights of persons with disabilities, social inclusion, and cultural participation, and are aligned with global frameworks such as the United Nations Convention on the Rights of Persons with Disabilities (CRPD). International joint R&D enables this project to go beyond technical development and contribute to the establishment of global best practices for guaranteeing cultural and religious participation rights for persons with disabilities.
Key Words	1) Sensory-Adaptive VR Worship Platform 2) AI-Based Personalized Worship Support Technology

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	3) AAC-Integrated Interactive System

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**R&D Objectives** 1) Development of VR Worship Environment Technologies Based on Diverse Religious Spaces ● Develop VR worship environments based on various religious spaces such as churches, cathedrals, and temples. The system constructs 3D virtual environments reflecting the structure and key elements of real worship spaces, while implementing technologies that allow stepwise adjustment of visual and auditory stimuli in consideration of the sensory characteristics of individuals with developmental disabilities. 2) Design and Implementation of Game-Based Worship Learning Scenarios ● Develop technologies that structure worship procedures into learnable units through game-based scenarios. By organizing the flow of worship into step-by-step missions and interactive elements, users can naturally acquire worship sequences and social norms through repeated experiences. 3) Development of AAC-Based Communication Support Technologies ● Develop AAC-based communication support technologies that simultaneously assist understanding and expression for individuals with developmental disabilities. Core worship-related concepts and actions are provided using AAC symbols, images, and simplified sentences, and interfaces are implemented to allow users to express intentions and make selections using AAC within the virtual environment. 4) Development of a Ubiquitous Learning Platform Considering Diverse Usage Environments ● Develop platform technologies that enable the same learning content to be used across multiple devices, including VR HMDs, tablets, and PCs. This allows continuous and repetitive learning in diverse contexts such as homes, educational institutions, and religious facilities. 5) Development of Personalized Learning and Sensory/Communication Configuration Technologies ● To maximize learning effectiveness according to individual user characteristics, develop technologies that allow personalized configuration and storage of sensory stimulus levels, AAC expression complexity, and learning stages, enabling adaptation to each user's developmental level and need.			

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R&D Target Specifications

System-Level Target Specifications

1) Platform Type
   - Immersive VR–based worship support platform for individuals with developmental disabilities
   - Supports both standalone HMD and PC-based VR environments
2) Operational Architecture
   - Hybrid architecture combining on-device VR execution with cloud-based content and model updates
3) Scalability
   - Ability to expand worship content, personalization models, and language/cultural settings without hardware changes

Core Functional Target Specifications

1) Sensory Modulation Functions
   - Stepwise control of visual stimuli (color contrast, brightness, motion density)
   - User-customizable auditory stimuli (volume, background sound suppression, speech clarity)
   - Immediate environment simplification mechanisms in response to sensory overload
2) Personalized Cognitive Adaptation
   - Adjustment of worship flow pacing based on the user's cognitive level and attention sustainability
   - Provision of a predictable worship structure (start → praise → message → prayer → closing)
   - Support for repetition, simplification, and visually symbolic–centered design
3) Communication and Participation Support (AAC-Integrated Interaction)
   - Worship participation through symbol-, icon-, and choice-based interfaces
   - AAC-integrated prayer and response expression functions for non-verbal users
   - Minimal-movement interaction modalities, including gaze, simple gestures, and button selection.

Key Performance Indicators	Target Value	Weight	Measurement & Verification Method
Worship Participation Level	≥ 30% improvement	25%	Pre–post test comparison
Understanding of Worship Procedures	≥ 30% improvement	20%	Pre–post test comparison
Reduction in Worship-Related Anxiety	≥ 25% reduction	20%	Behavioral observation checklists and evaluations by caregivers and professionals
speech, Language, Communication, and AAC Performance	≥ 40% improvement	20%	Speech/language/AAC assessments, virtual environment log analysis, and real-world observation

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R&D Plan	**Research Approach & Strategy** 1) Participatory Research Approach ● Conduct participatory research involving individuals with developmental disabilities, caregivers, AAC specialists, and religious practitioners to ensure that real-world needs are reflected in the design. This approach secures practical applicability and user acceptance from the early stages of technology development. 2) Data-Driven Decision-Making Strategy ● Analyze user behavior data, learning performance outcomes, and AAC usage patterns collected throughout the research process, and incorporate the findings into functional improvements and system refinement. By adopting an iterative improvement cycle based on quantitative indicators rather than subjective judgment, the research ensures objectivity and verifiability. 3) Development of an Inclusive Worship Learning Technology Model ● Through international collaborative research, validate applicability across diverse cultural and religious contexts, and strengthen the universality and scalability of the technology design. **Year 1: Requirements Analysis and Initial Prototype Development** 1) Core Objective ● Requirements analysis and completion of the first prototype 2) Major Tasks ● Analyze worship participation characteristics, sensory responses, and AAC usage levels of individuals with developmental disabilities ● Analyze the structure and procedures of worship spaces such as churches, cathedrals, and temples ● Design the foundational game-based learning scenarios by structuring worship procedures into stages ● Design symbol, image, and simplified sentence systems to support AAC-based understanding 3) Expected Milestones ● User requirements analysis report ● First prototype of the virtual worship environment **Year 2: Functional Advancement and Pilot Validation through Integrated Systems** 1) Core Objective ● Functional advancement and pilot validation through integration of game-based learning, AAC interfaces, and a ubiquitous learning environment 2) Major Tasks ● Advance game-based worship learning scenarios and enhance interactive elements ● Implement integrated AAC-based understanding and expression interfaces and

	develop personalization functions ● Analyze user behavior data and learning performance outcomes 3) Expected Milestones ● Beta version of the integrated worship learning system ● AAC-integrated game-based learning content ● Implementation of personalized configuration and data collection functions **Year 3: Performance Validation through International Demonstration and Final System Completion** 1) Core Objective ● Performance validation through international joint demonstrations, completion of the final system, and preparation for dissemination 2) Major Tasks ● Conduct international joint demonstration studies in multi-national and multi-cultural environments ● Expand application scenarios across different religious traditions ● Consolidate research outcomes and review standardization potential 3) Expected Milestones ● Final integrated worship learning platform

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Conceptual Design and Theoretical Framework Development** 1) Technical Contributions and Research Scope ● Analysis of worship participation characteristics and AAC usage patterns of individuals with developmental disabilities ● Establishment of user experience (UX) and interaction design principles for VR-based worship learning environments ● Research validation of game-based learning–structured AAC integration models ● Derivation of universal accessibility design criteria applicable across multicultural and multi-religious environments 2) Specific Tasks ● Conduct user requirements analysis and define core AAC-based design requirements ● Perform theoretical validation of virtual worship environments and learning scenarios ● Participate in the design and analysis of international joint pilot and demonstration studies ● Systematically organize research outcomes and propose directions for technology advancement 3) Provided Resources ● Specialized research personnel in computer science and human–computer interaction (HCI) ● Support for user evaluation design and research methodologies ● Provision of analysis frameworks and data interpretation support for international joint demonstration studies
Expected R&R of Korean SME Partner	**Applied Technology Commercialization and Production Validation** 1) Technical Contributions and Development Tasks ● Development of 3D content for VR-based worship environments ● System implementation of game-based worship learning scenarios ● Development of AAC-based interfaces and personalized user functions ● Construction of device integration and platform infrastructure supporting a ubiquitous learning environment 2) Testing and Validation Activities ● Implementation of virtual worship environments and game-based learning content ● Technical development of AAC interfaces and personalization functions ● System integration, stabilization, and performance optimization ● Execution of domestic pilot testing and incorporation of technical improvements ● Presentation of technology application models for commercialization and dissemination 3) Collaboration Approach and Resources ● VR content development personnel and software development personnel ● Development environments for system integration and testing ● Technical support resources for pilot testing and field deployment

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5. Expected Outcomes/Impact (기대 성과 및 파급 효과)

Expected Outcomes and Impact	**Technical Outcomes** ● Implementation of a VR worship environment that dynamically adjusts visual, auditory, and interaction elements in real time based on the cognitive levels and sensory characteristics of individuals with developmental disabilities. ● Development of AI-based personalization models grounded in cognitive and sensory processing theories, ensuring stable operation and generalization performance even in environments with high inter-user variability. ● Application of AAC-integrated interaction technologies that enable meaningful worship participation—such as prayer, responses, and choice-making—for users with limited verbal expression. ● Establishment of a cloud-based remote update framework for content and personalization models, ensuring continuous scalability and efficient maintenance. ● Definition of international-level design principles and evaluation metrics for worship environments tailored to individuals with developmental disabilities. **Commercialization Potential** ● Primary target markets: churches and religious institutions, welfare facilities for individuals with developmental disabilities, special education institutions, and public religious and cultural service sectors. ● Business models: cloud-based subscription (SaaS) services, denomination- or institution-specific content and functionality licensing, and strategic partnerships with domestic and international religious and welfare organizations. ● Scalability: applicability across diverse sites without requiring hardware modifications or specialized personnel, enabling easy domestic and international dissemination. ● Foundation for global market entry: reduced barriers to global expansion through credibility established via overseas demonstrations and international collaboration. **Industrial and Economic Impact** ● Expansion of XR technology applications from entertainment- and education-centered domains to inclusive religious and social service sectors. ● Strengthening national competitiveness in the global assistive and inclusive technology market, which is growing in response to aging populations and expanding disability inclusion policies. ● Creation of social value and reinforcement of technology-enabled public good by expanding opportunities for religious and community participation among individuals with developmental disabilities. ● Spillover effects into adjacent industries, including special education, rehabilitation, mental health, and public cultural services. ● Establishment of an international reference model leading inclusive digital transformation.

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6. References

(1) Parsons, S., & Cobb, S. (2011). State-of-the-art of virtual reality technologies for children on the autism spectrum. *European Journal of Special Needs Education*, 26(3), 355–366. https://doi.org/10.1080/08856257.2011.593831

(2) Bekele, E., Wade, J., Bian, D., Fan, J., Swanson, A., Warren, Z., & Sarkar, N. (2016). Multimodal adaptive social interaction in virtual environment (MASI-VR) for children with autism spectrum disorders. *Proceedings of the IEEE Virtual Reality (VR)*, 121–130. https://doi.org/10.1109/VR.2016.7504695